

Amador M. Guzmán · Marcelo Del Valle

Heat transfer enhancement in grooved channels due to flow bifurcations

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Abstract The flow bifurcation scenario and heat transfer characteristics in grooved channels, are investigated by direct numerical simulations of the mass, momentum and energy equations, using the spectral element methods for increasing Reynolds numbers in the laminar and transitional regimes. The Eulerian flow characteristics show a transition scenario of two Hopf bifurcations when the flow evolves from a laminar to a time-dependent periodic and then to a quasi-periodic flow. The first Hopf bifurcation occurs to a critical Reynolds number Re_1 that is significantly lower than the critical Reynolds number for a plane-channel flow. The periodic and quasi-periodic flows are characterized by fundamental frequencies ω_1 and $m \cdot \omega_1 + n \cdot \omega_2$, respectively, with m and n integers. Friction factor and pumping power evaluations demonstrate that these parameters are much higher than the plane channel values. The time-average mean Nusselt number remains mostly constant in the laminar regime and continuously increases in the transitional regime. The rate of increase of this Nusselt number is higher for a quasi-periodic than for a periodic flow regime. This higher rate originates because better flow mixing develops in quasi-periodic flow regimes. The flow bifurcation scenario occurring in grooved channels is similar to the Ruelle-Takens-Newhouse transition scenario of Eulerian chaos, observed in symmetric and asymmetric wavy channels.

Nomenclature

A	Area
H	Separation between walls
L	Streamwise separation among blocks
P	Pressure

$\dot{Q}$	Volumetric flow rate
T	Temperature
Pr	Prandtl number
Re	Reynolds number
Re_c	Critical Reynolds number
h	Height of the block
k	Thermal conductivity
l	Length of the block
n, m	Integer numbers
$\overline{Nu}$	Global Nusselt number
$Nu(x)$	Local Nusselt number
$q(x)$	Local heat flux
$T_w(x)$	Local lower wall temperature
T_b	Bulk temperature
ΔP	Pressure drop
$\hat{U}$	Time-average mean streamwise velocity
$\vec{v}$	Velocity
$\dot{q}'''$	Heat generation
c_p	Specific heat
$\tilde{h}$	Half height of the channel
$\tilde{h}$	Characteristic length
f_b	Block tandem channel friction factor
f_p	Plane channel friction factor

Greek symbols

α	Thermal diffusivity
μ	Dynamic viscosity
ν	Kinematic viscosity
ρ	Density
ω	Fundamental (critical) frequency

A. M. Guzmán (✉) · M. D. Valle
 Departamento de Ingeniería Mecánica,
 Universidad de Santiago de Chile,
 3363 Alameda, Santiago, Chile
 E-mail: aguzman@lauca.usach.cl
 Tel.: +56-2-7786381

1 Introduction

A significant amount of research has been performed in the last decades to obtain a better understanding of flow mixing and heat transfer enhancement in channels with

geometrical inhomogeneities, such as grooved, communicating, wavy and corrugated channels, and other related geometries, such as backward facing steps, channel expansions, passages with eddy promoters, and grooved tubes [1–9]. Grooved channels are used in a whole range of industrial and biomedical applications, such as compact heat exchangers and cooling of electronic equipment, and oxygenators and dialysers. Researchers have investigated several active and passive heat transfer enhancement schemes in an effort to enhance the heat transfer rates by affecting flow characteristics. Considerable effort has been dedicated to investigating the role of flow destabilization and flow mixing in heat transfer enhancement. Active flow modulation for example, has been shown to be effective in producing resonant heat transfer enhancement. Several investigations performed with grooved, communicating and wavy channels, have demonstrated that self-sustained oscillations that develop in these geometries by exciting flow instabilities, lead to heat transfer enhancement without the need to apply active forcing. It has been found that amounts of heat transfer enhancement vary in a wide range, depending on the specific geometric characteristics and boundary conditions.

The configuration of computer chips on a motherboard can be seen as a channel with a tandem of blocks, or as a grooved channel with many rectangular grooves on one wall. In the block tandem channel, heat generated within each block must be removed efficiently with a minimum dissipation penalty to maintain block temperature at a reasonable level. Several active and passive techniques of flow destabilization have been used in different grooved channel configurations to enhance the flow mixing and heat transfer processes. Experimental, analytical and numerical investigations have been performed by Ghaddar et al. [1], Amon and Mikic [2], Greiner [3], Greiner et al. [10–12], Pereira and Sousa [13], Farhanieh et al. [14], Nigen and Amon [4], Nishimura et al. [5, 15], and Wirtz et al. [16], to investigate the laminar and transitional flow pattern and heat transfer characteristics. These studies have determined that the laminar flow undergoes a flow bifurcation at a critical Reynolds number that is significantly lower than the critical Reynolds number for a Poiseuille plane channel flow. These investigations indicated that the heat transfer rate increases when the flow passes from a laminar to a transitional regime, due to better flow mixing after flow bifurcation, and that usually, this enhancement is accompanied by an increase in the pressure drop [2, 5–17].

In recent studies Nishimura et al. [15], experimentally investigated the influence of the imposed oscillatory frequency on mass transfer enhancement of grooved channels with different cavity lengths, for an external pulsatile flow. It was found that the mass transfer enhancement by means of fluid oscillations is higher in laminar than in turbulent flow and there is noticeable enhancement at intermediate Strouhal numbers, depending on the cavity length and Reynolds number.

In previous experimental studies of fluid mixing in grooved channels for a constant flow rate, Nishimura et al. [5] found self-sustained oscillations whose onset depends on the length of the cavity. Recently, Adache and Uehara [17] performed numerical investigations of flow and temperature fields for steady state and self-sustained oscillatory flows in periodically grooved channels for various channel geometrical configurations. They determined correlations between heat transfer and pressure drops and they found that heat transfer is significantly enhanced, and pressure drop increases after the first flow bifurcation. They also determined that expanded grooved channels perform more efficiently than contracted grooved channels. Recently, Herman and Kang [6] investigated the heat transfer enhancement in a grooved channel with curved vanes by using holographic interferometry. They found that flow oscillations develop when the flow evolves from a laminar to a transitional regime and heat transfer rates increase by a factor of 1.5–3.5 when compared to the basic grooved channel.

Several investigations have determined that a natural transition to a time-dependent, periodic self-sustained flow regime, lead to enhancement of heat transfer rates, with moderate and reasonable pressure drops. Guzmán and Amon [7, 18, 19] performed numerical investigations for high transitional Reynolds numbers in converging-diverging (symmetric wavy wall) channels, and they found that the flow develops a second bifurcation, in addition to the first flow bifurcation at a Reynolds number $Re_{c2} > Re_{c1}$, which enhances flow mixing and heat transfer. Experimental or numerical investigations of the flow pattern and heat transfer characteristics for higher transitional Reynolds numbers for this type of grooved channels have so far not been reported. This paper reports numerical investigations of the transition scenarios and heat transfer characteristics in grooved channels as the flow regime evolves from a laminar to a transitional regime by two-dimensional direct numerical simulations (DNS) of the time dependent, incompressible mass, Navier–Stokes and energy equations. The flow bifurcation scenario is described here from a laminar to a transitional state; flow and heat transfer characteristics and parameters are described through the transitional regime for increasing Reynolds numbers.

2 Problem definition

The physical configuration of the channel used in this investigation is shown in Fig. 1. The grooved channel has an upper flat wall and a lower wall with a tandem of several blocks. The distance between both walls is H . Each block has a height h , a length l , and a separation among blocks given by the streamwise distance L . The channel periodic length is $(L + l)$ and the flow is from left to right. Heat is generated within each block in a very small region, as shown in this figure. This investigation uses extended computational domains to

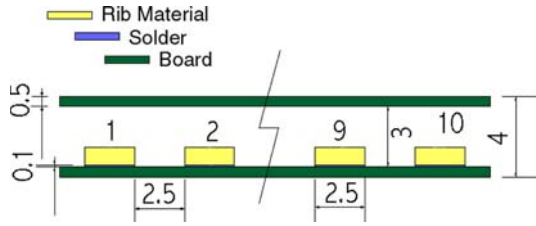


Fig. 1 Geometrical characteristics in a grooved channel

determine the existence of a fully developed flow and self-similar temperature profiles, and then uses reduced computational domains to investigate the flow characteristics and heat transfer enhancement for laminar and transitional regimes. Natural convection effects are not considered here as a calculation of Gr/Re^2 based on typical experimental and numerical simulations data in the current configuration and in similar geometries yields a value much smaller than one [18, 26–30]. The transition scenario and associated heat transfer process are investigated in detail with the reduced domain models. This article reports numerical simulation results within the reduced domain.

3 Numerical approach and methodology

Unsteady and incompressible flows of a Newtonian fluid, for both the extended and reduced domains, are considered here. Flow is governed by the mass conservation and Navier–Stokes equations Eq. 1, Eq. 2; whereas, the heat transport process is described by the energy equation Eq. 3. For the extended domain model, a parabolic velocity profile was imposed at the entrance of the channel—non-slip for the upper and lower and block walls and outflow for the channel exit.

To address the heat transfer problem, a constant temperature was imposed at the entrance, a vanishing heat flux on the upper wall, outflow for the exit, and a constant heat generation of $4,444 \text{ W/m}^3$ within each block of the grooved channel. For the reduced domain, periodicity was applied in the streamwise direction, which corresponds to a fully developed flow. For the heat transfer problem, a self-similar temperature profile was enforced by imposing periodicity in the streamwise direction. As in the extended domain, heat is generated within a small region of the block, which represents the heat source, as shown in Fig. 2.

$$\nabla \cdot \vec{V} = 0 \quad (1)$$

$$\frac{\partial \vec{V}}{\partial t} + \vec{V} \cdot \nabla \vec{V} = -\frac{1}{\rho} \nabla P + \frac{\mu}{\rho} \nabla^2 \vec{V} \quad (2)$$

$$\rho c_p \left(\frac{\partial T}{\partial t} + \vec{V} \cdot \nabla T \right) = k \nabla^2 T + q''' \quad (3)$$

The governing equations are solved using a computational program based on the spectral element method

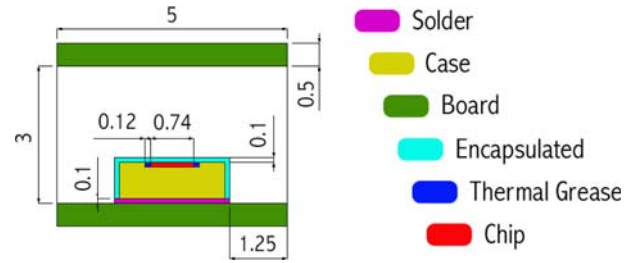


Fig. 2 Computational reduced domain of the grooved channel

(SEM), which combines the high accuracy of the spectral methods with the geometric flexibility of the finite element method [20]. The computational domains are discretized with macroelements containing nodal points in which velocity, pressure and temperature are represented. To obtain good spatial and temporal resolutions of both velocity and temperature, macroelements of smaller sizes were used near the walls, where steep velocity and temperature gradients are expected. A mesh refinement study with different discretizations and mesh resolutions was performed to establish the adequacy of the computational discretization and the imposed boundary conditions. Comparisons with available experimental information were performed to verify and validate the periodicity assumptions for flow and temperature. Figure 3 shows the computational mesh and discretization for the grooved channel reduced domain. Notice the finer discretization near the block for the grooved channel and the regular discretization elsewhere.

Numerical simulations were performed for increasing Reynolds numbers in the laminar and transitional flow regime. The procedure for obtaining asymptotic stable states for a given Reynolds number—and consequently a volumetric flow rate—consists of integrating the governing equations in time, starting with a predicted steady flow and gradually increasing the Reynolds number until a steady, time periodic, or transitional state is found. Once an asymptotic stable state is obtained, the governing equations for a different Reynolds number are solved using the previous asymptotic state as the initial condition. For this investigation, the Reynolds number is defined as

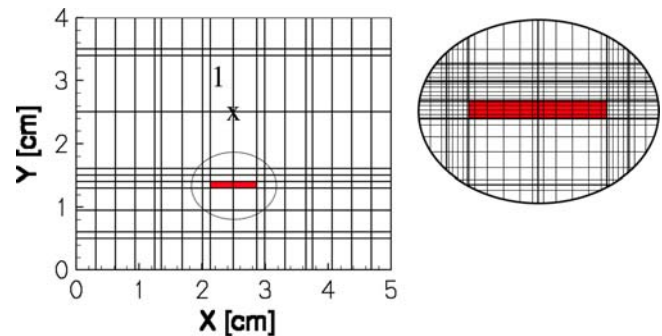


Fig. 3 Domain discretization for the grooved channel

$$Re = \frac{3 \hat{U} \cdot \hat{h}}{2 \nu}$$

where $\hat{U}$ is the time-average mean streamwise velocity at the inlet of the channel, which is calculated as Q/A , with Q and A , the volumetric flow rate and the cross-sectional area, respectively; $\hat{h}$ is half-height of the channel at the inlet, and ν is the kinematic viscosity. The Prandtl number, $Pr = \nu/\alpha$ is equal to 1, and α is the thermal diffusivity. Since the fluid motion is induced by an external applied pressure drop, and not for body forces, the resulting flow pattern and characteristics are independent of the imposed thermal boundary conditions and temperature distribution.

Flow and heat transfer simulations were performed for the Reynolds number range $Re = (0, 905)$, which corresponds to laminar and transitional flow regimes. Figure 4 shows the streamwise u -velocity as a function of time of a typical point of the domain for laminar Reynolds numbers. After the flow becomes completely steady, at a time of about 25 s, the flow evolves to another steady state after an initial transient state because of an increase in the volumetric flow rate. In some situations, this higher laminar Reynolds flow regime was used as the initial condition for a lower (the initial) volumetric flow rate, with the objective of determining hysteretic effects, which are important in order to determine and build the correct bifurcation diagram. No hysteretic effects were detected in the range of the laminar flow simulations.

4 Numerical results and discussion

4.1 Laminar and transitional flow regimes

This section presents numerical results obtained with the reduced domain models for laminar and transitional flow regimes. The Reynolds number range for a laminar regime is approximately $Re = 1$ –545. Figure 5 shows streamlines for a laminar Reynolds number of $Re = 542$. This figure shows that the flow pattern is characterized by a parallel flow in the mid-section and a big stationary

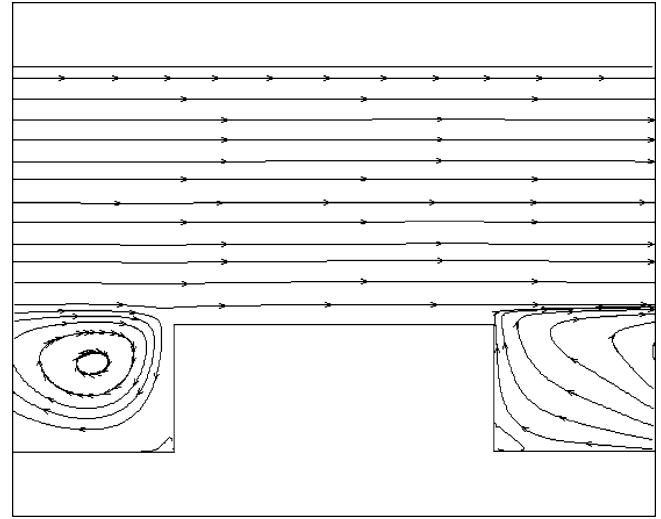


Fig. 5 Streamlines for a laminar Reynolds number, $Re = 542$

vortex in the grooved region between blocks. This flow pattern changes when the flow evolves to a transitional periodic regime through a first Hopf bifurcation. The periodic flow presents one fundamental frequency ω_1 , and super-harmonics $2\omega_1, 3\omega_1, 4\omega_1, \dots, n\omega_1$, with n , an integer number.

Figure 6 shows a sequence of six instantaneous streamline representations during one τ time period for a periodic transitional Reynolds number of $Re = 549$. These sequences demonstrate the wavy periodic nature of the flow as it moves downstream, and it shows a flow pattern of several vortices of different sizes, which increase and decrease in size and are ejected from the cavity to the mean flow. The flow pattern for higher transitional Reynolds numbers preserves its wavy nature; however the vortex dynamics increases in intensity. Figure 7 shows the Eulerian flow characteristics for a quasi-periodic flow regime in the grooved channel, in terms of temporal evolution of the u -velocity component, Fourier power spectra and phase-portrait of the velocity components in a characteristic point of the computational domain indicated as 1 in Fig. 3. These representations show that the flow has evolved to a quasi-periodic flow regime from a periodic flow, through a second Hopf bifurcation at the critical Reynolds number of $Re_{c2} \approx 860$. The Fourier power spectra show that the flow presents two fundamental frequencies ω_1 and ω_2 , and linear combination of these frequencies $m\omega_1 + n\omega_2$, where m and n are integers. The phase portrait depicts the quasi-periodic flow nature, where a continuous shifting in the phase portrait representation develops because of the existence of the super harmonics $m\omega_1 + n\omega_2$. The fundamental frequency ω_2 , obtained in this quasi-periodic flow, is related to the frequency that a fluid particle, traveling with the flow to a time-average mean streamwise $\hat{U}$ velocity, encounters or reaches a downstream spatially periodic geometric perturbation. These perturbations (or geometric inhomogeneities) are separated by the streamwise length L . The relationship

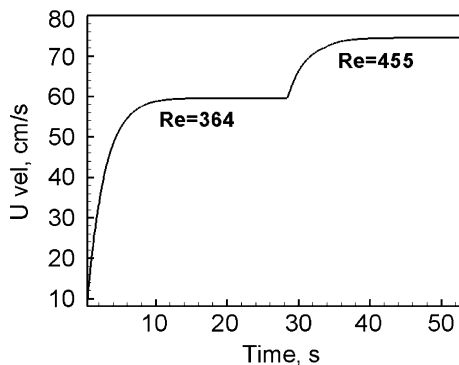
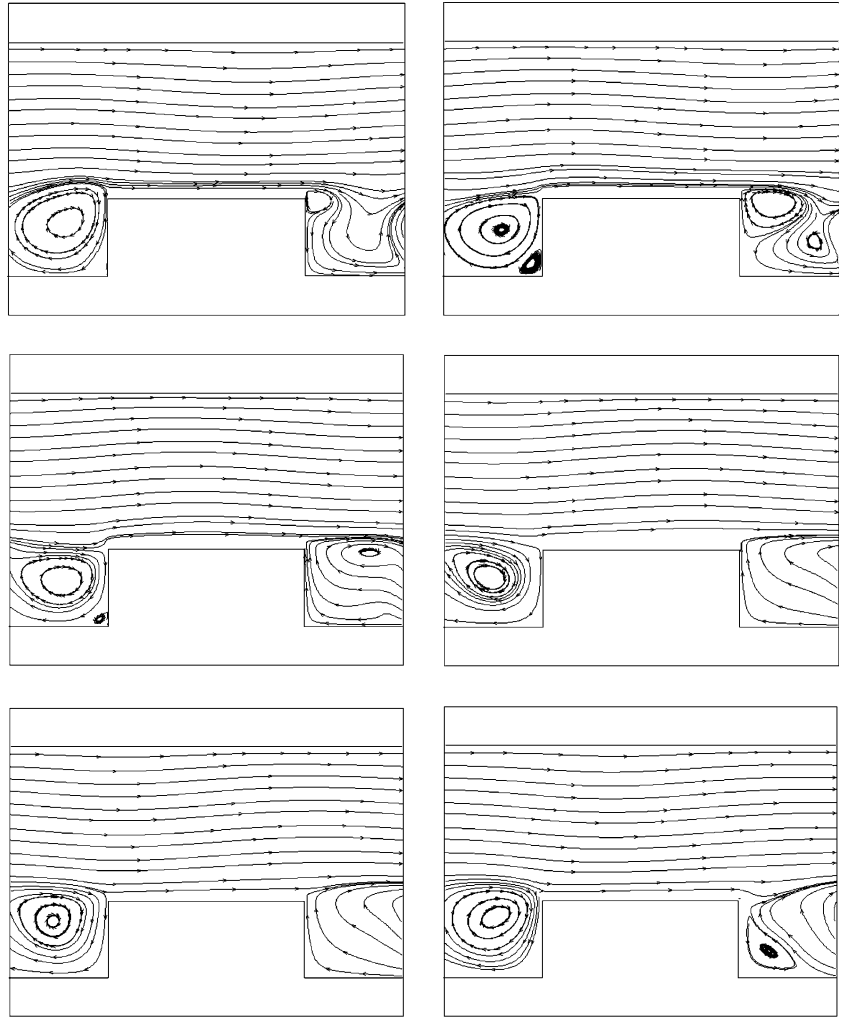


Fig. 4 U -velocity temporal evolution between two steady states

Fig. 6 Sequence of six instantaneous representations of streamlines during one time period for a transitional flow regime of $Re = 549$



between the second fundamental frequency ω_2 and the geometric perturbations is being currently investigated and tested in other channel configurations.

Figure 8 shows a schematic representation of the transition scenario from a laminar to a quasi-periodic flow by two successive Hopf bifurcations. These Hopf bifurcations occur for Reynolds numbers Rec_1 and Rec_2 , which are significantly lower than the critical Reynolds number for a Poiseuille plane channel flow. In the grooved channel, the flow loses stability at critical Reynolds numbers Rec_1 and Rec_2 , which are lower than the critical Reynolds number for the plane Poiseuille channel. This transition scenario is similar to the Ruelle-Takens-Newhouse scenario to chaos developed in converging-diverging channels [18].

This transition scenario from a steady state to periodic flow and then to a quasi-periodic flow, generates a flow pattern characterized by high velocity and velocity gradients, which generate high shear stresses on the block surface of the grooved channel and, consequently, a high friction factor. Figure 9 shows the friction factor as a function of the Reynolds number for both the grooved channel and the Poiseuille plane channel. The grooved channel friction factor is calculated as

$$f_b = \frac{-(dp/dx) \cdot 8\hat{h}}{2\rho\hat{U}^2},$$

where $dp/dx = \Delta P/\hat{L}$ is the fully developed pressure gradient in the streamwise direction; and, $\hat{L} = L + 1$. The Poiseuille plane channel friction factor is calculated as $f_p = 18/Re_H$, where Re_H is the Reynolds number based on the height H of the plane channel. As expected, the friction factor for the grooved channels is higher than the friction factor for the Poiseuille plane channel for the Reynolds number range of this investigation.

As shown in Fig. 9, the friction factor curve shows that during the laminar regime, the slopes for the grooved and plane channel curves are very similar up to a Reynolds number close to the critical Reynolds $Rec_1 \approx 549$. The mean flow in the grooved channel is parallel to the streamwise direction, with one stationary vortex and recirculation region in the cavity between blocks, which increases the pressure drop with respect to the plane channel, and consequently raises the friction factor. In the periodic flow regime, between the two Hopf bifurcations at Rec_1 and Rec_2 , the slope of the friction factor for the grooved channel increases with respect to the Poiseuille plane channel friction factor. This increase is caused by

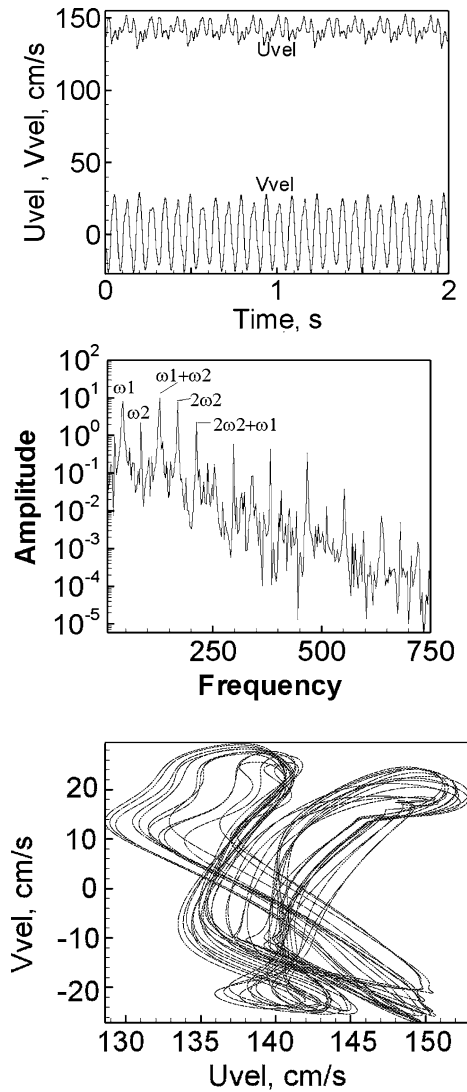


Fig. 7 Temporal evolution of the U -velocity, Fourier power spectra of the U -velocity, and phase-portrait of the U - and V -velocities for a quasi-periodic flow ($Re = 860$)

the stronger vortex dynamics and better flow mixing developed in this transitional flow regime. The flow in the grooved channel loses a higher amount of energy than the Poiseuille plane channel because in the grooved channel the flow changes direction to fill the cavity between blocks and push the fluid out of the channel.

4.2 Heat transfer characteristics

The temperature distribution in the grooved channel is shown in Fig. 10 for laminar and transitional flow

Fig. 8 Transition scenario from a laminar to a quasi periodic flow by two successive Hopf bifurcations, B_1 and B_2 for the critical Reynolds numbers $Rec_1 = 549$ and $Rec_2 = 860$, respectively

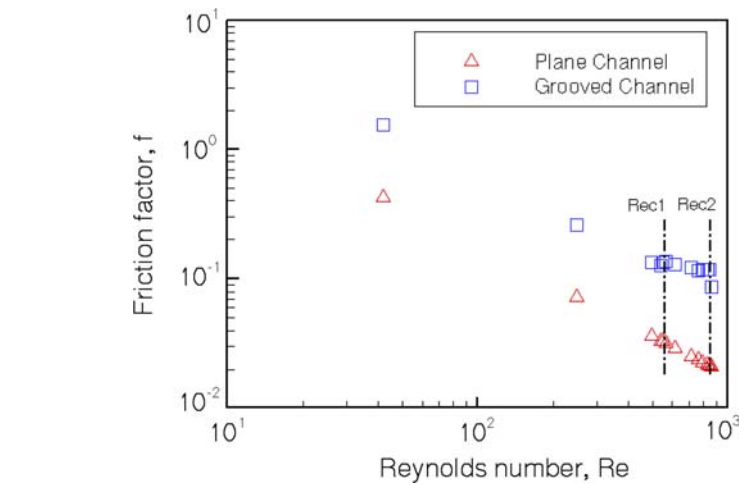


Fig. 9 Grooved and plane channels friction factors versus Reynolds number

regimes. Figure 10a shows a thermal stratification for the laminar Reynolds number of $Re = 542$. The heat transfer process is dominated by heat diffusion in the crosswise direction and a combination of diffusion and convection in the streamwise direction. The block and fluid within the cavity in the grooved channel seem to present a stratified temperature distribution. A close-up of the block shows, however, that the highest temperature is located in the heat source and decreases to the external block surfaces. The temperature distribution within the block is due to the higher thermal conductivity of the components of the block that allows the heat to be removed from the heat source to other regions. Because of the lack of flow mixing and the existence of stationary recirculation regions in the grooves, the heat is not transported to the mean flow or to the upper wall where the temperature is lower than near the lower wall. Figure 10b shows six (6) instantaneous representations of the temperature distribution within the grooved channel during one period of time τ for a periodic flow regime of $Re = 549$. Because of the improved flow mixing and the stronger vortex dynamics, hot fluid from the block surface is projected to the mean flow and to the exit of the channel. At the same time, the hot fluid in the front of the block is ejected onto the upper surface of the block and the mean flow, disrupting the thermal boundary layer on the block surface and increasing the transport of hot fluid to regions with a low temperature. The better flow mixing also allows the transportation of fluid from the upper surface, with low temperature, to a region with a higher temperature close

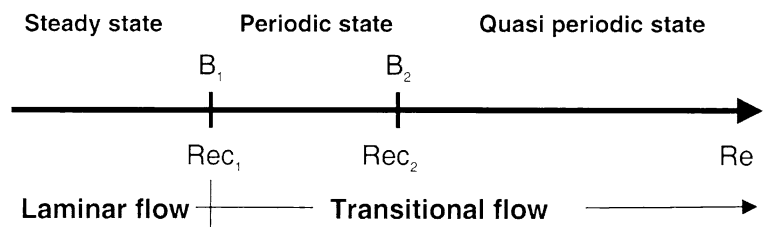
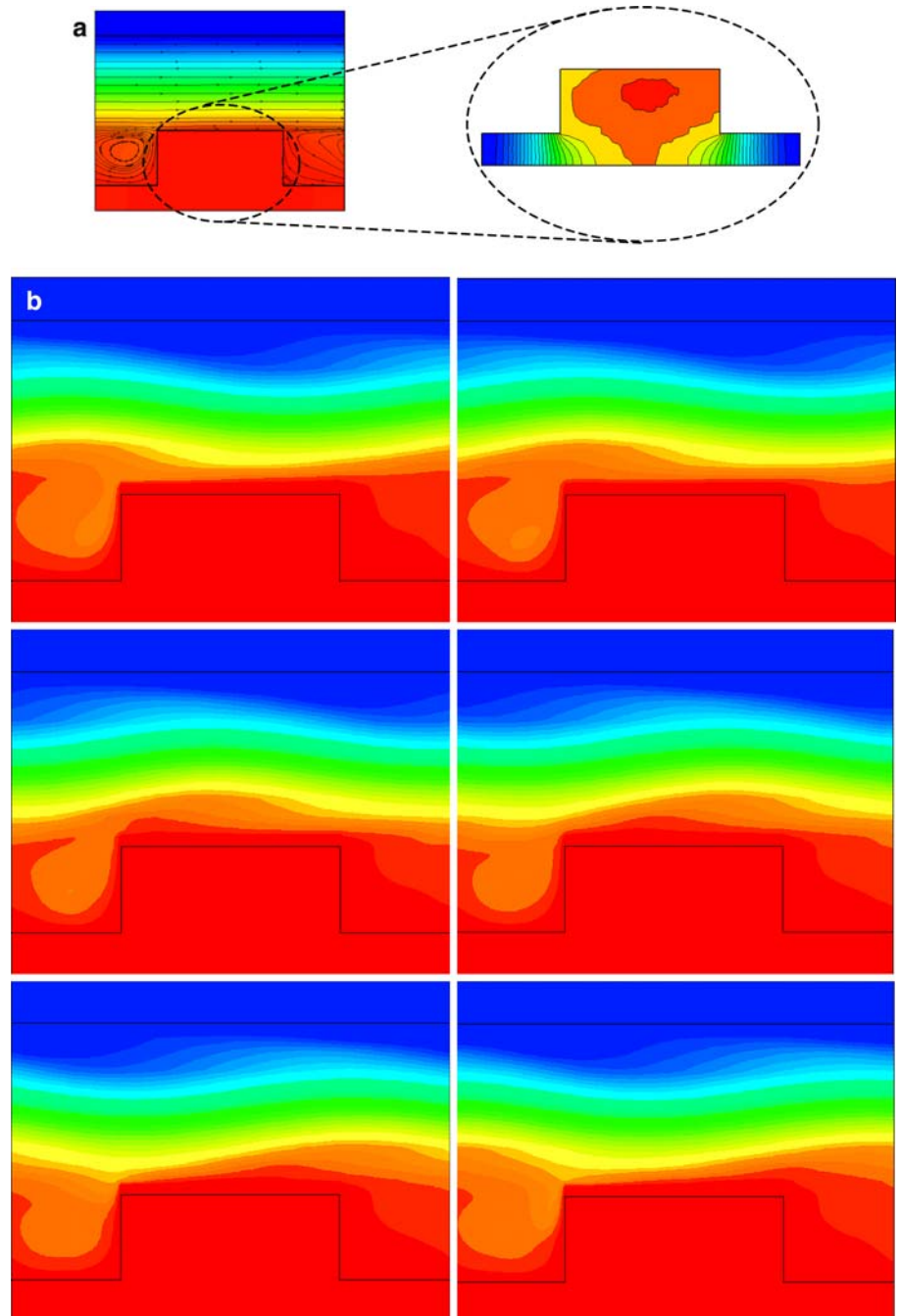


Fig. 10 Temperature distribution: **a** laminar flow ($Re = 542$); and, **b** instantaneous representations of a sequence six times for a periodic flow ($Re = 549$)



to the block, thus cooling the block surface. Thus, this periodic transitional flow regime enhances the heat transport from the inner region of the block to the surface of the block and lowers the inner block temperature. The time-dependent temperature field for this periodic flow mimics the wavy nature and vortex dynamics of the flow field. This results shows that the flow mixing occurring in this transitional flow causes an enhancement in the heat transfer rates. Moreover, for a quasi-periodic flow, the flow mixing enhancement (not shown here) increases even more the heat transfer rates, as will be indicated later.

Figure 11 shows the time-average mean Nusselt number as a function of the Reynolds number for laminar and transitional flow regimes. The mean Nusselt number is calculated as

$$\overline{Nu} = \frac{\int \hat{L} Nu(x) dx}{\int \hat{L} dx}$$

where $Nu(x)$ is the local Nusselt number evaluated at the lower wall $Nu(x)|_{wall} = q(x) \cdot \tilde{h}/k \cdot \langle T_w - T_b \rangle$ and $\hat{L}$ is the integration length; $q(x)$ is the local heat flux along the

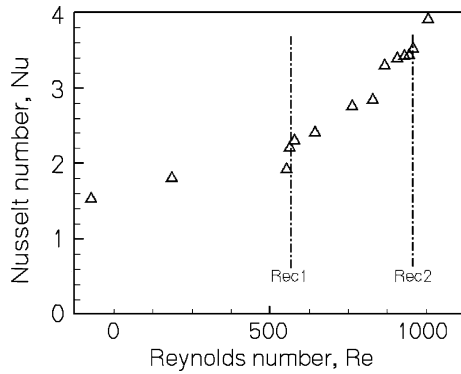


Fig. 11 Time-average mean Nusselt number versus Reynolds number for laminar and transitional flow regimes. The Hopf bifurcations B_1 and B_2 , occur at the critical Reynolds number Rec_1 and Rec_2 , respectively

lower wall of the grooved channel, $\tilde{h}$ is the characteristics length, $T_w(x)$ is the lower wall temperature and T_b is the bulk temperature defined as

$$T_b = \frac{\int_a^b u(x_o, y) \cdot T(x_o, y) \cdot dy}{\int_a^b u(x_o, y) \cdot dy}$$

where x_o represents the channel inlet, and a and b are the lower and upper walls. The mean Nusselt number $\bar{Nu}$ represents the instantaneous Nusselt number, which is time-independent for a laminar flow. For a transitional flow regime, the mean Nusselt number is integrated over a period of time and averaged for the total time associated with the time period.

The time-average mean Nusselt number—or simply the Nusselt number—increases with the Reynolds number as the flow evolves from a laminar to a transitional regime. The Nusselt number increases slightly in the laminar regime, up to a Reynolds number $Re < Rec_1 \approx 549$. In this regime, the rate of increase of the Nusselt number is small because the heat transport to the mean flow is mainly by crosswise diffusion. As the flow evolves to a time-periodic regime, both the Nusselt number and the rate of increase of the Nusselt number increase farther. These increases originate with improved flow mixing, which develops in the periodic flow, with momentum and heat fluxes being transported away from the hot walls into the mean flow. The Nusselt number has increased for at least a factor of two, with respect to the laminar Nusselt number. Last, the Nusselt number, and the rate of increase of the Nusselt number, increases even more as the flow passes the critical Reynolds number Rec_2 (associated with the second Hopf bifurcation), and becomes quasi-periodic. During this quasi periodic regime, the flow develops a stronger flow mixing and vortex dynamics, which enhance the heat transfer process. The Nusselt number has increased by a factor of ≈ 2.5 in the grooved channel with respect to the laminar regime.

5 Summary and concluding remarks

A transition scenario of two Hopf bifurcations develops in the grooved channel as the Reynolds number increases from a laminar to transitional regime. The first bifurcation at a $Re = Rec_1$ originates a periodic flow characterized by one fundamental frequency ω_1 . The second Hopf bifurcation at a $Re = Rec_2 > Rec_1$, originates a quasi-periodic flow with two fundamental frequencies ω_1 and ω_2 and linear combinations of these frequencies. The laminar flow is characterized by a parallel mean flow and stationary vortices. In the transitional regime the periodic and quasi periodic flows are self-sustained and wavy in nature, with active vortex dynamics and enhancement of flow mixing. The friction factor is higher than the Poiseuille plane channel friction factor. The slope of the friction factor changes as the flow evolves from a laminar to transitional regime. In the laminar regime, the slope of this curve is close to the Poiseuille plane channel; whereas, in the transitional regime, and after the first bifurcation, the slope of the grooved channels friction factor curve becomes larger.

For the laminar regime, the temperature distribution shows a pattern of stratification with the isotherms parallel to the streamlines; whereas, in the transitional regime, the temperature distribution mimics the oscillatory wavy nature of the periodic and quasi-periodic transitional flows. The vortex dynamics in the transitional regime enhances the flow mixing and improve the heat transfer process from the hot surfaces to the mean flow by transporting hot fluid to cooler fluid regions, which enhances the heat transfer process. There is an enhancement of heat transfer as the Reynolds number increases from the laminar to transitional flow regimes. In the laminar regime the time-average mean Nusselt number remains almost constant; whereas, a significant heat transfer enhancement is obtained with supercritical transitional flow Reynolds numbers. This enhancement is by a factor of two in the periodic flow regime and a factor of 2.5 for the quasi-periodic regime. This enhancement is obtained without the necessity of operating this channel to high volumetric flow rates associated with turbulent flow regimes, which demand high pumping powers.

This investigation has demonstrated that, while no significant heat transfer enhancement is obtained for a laminar flow, significant heat transfer enhancements are obtained when the grooved channel is operated in the appropriated transitional Reynolds number range between both Hopf bifurcations and after the second Hopf bifurcation.

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